

A New Age Of Deception In Warfare

By Rand Waltzman

A ground-based missile defense system using machine learning to identify and intercept incoming threats is deployed to protect a strategic base. The system relies on radar signatures to classify objects as friendly, hostile, or non-threatening. Adversaries launch a coordinated attack by releasing a swarm of small drones programmed to emit radar signals that mimic incoming missiles. The defense system, overwhelmed by the sheer number of targets, prioritizes the drones, expending its missile interceptors. Amid the confusion, a genuine missile strike penetrates the base's defenses, causing significant damage.

While the scenario above is fictional, it represents a plausible future scenario. It helps to highlight how even sophisticated systems can be overwhelmed and rendered ineffective through deliberate deception, and underscores the critical need for robust validation mechanisms and human oversight. As we stand on the brink of a new technological era, the integration of machine learning and artificial intelligence into military systems introduces not only promise but also profound risks.

Behind this realization is a stark reality. Deception, an ancient tactic in warfare, is evolving. Where humans were once the primary targets of such operations, the rise of intelligent machines opens a new battlefield: deceiving technology itself. Unlike traditional cyberattacks, which often rely on breaching a system's defenses to access its internal data or software, deceiving machine learning systems requires no such intrusion. Instead, adversaries can manipulate these systems by exposing them to carefully crafted inputs through external sensors, such as cameras, microphones, or radar systems.

This approach mimics the way humans can be deceived—not by tampering with the brain directly but by presenting misleading or false information to the senses. For example, altering a stop sign's appearance to confuse an autonomous vehicle into misidentifying it as a speed limit sign relies entirely on manipulating what the vehicle "sees." This kind of attack exploits the system's reliance on patterns and inputs, demonstrating how deception has transcended its historical focus on human perception to target the very tools designed to assist us.

LEARNING BY EXAMPLE

The ability to learn by example is one of the most powerful and mysterious forces driving intelligence, both in humans and machines. Think of how children learn to recognize a letter or an animal. By being exposed to enough examples—whether it's the letter "B" or a cat—they effortlessly develop the ability to identify new instances of that letter or animal. This seemingly magical process stems from the brain's natural tendency to find patterns. It identifies the essential qualities of "cat-ness" or "B-ness," allowing the child to categorize experiences unconsciously and automatically. However, while this process feels intuitive, explaining it remains one of the great mysteries of intelligence. The issue isn't forgetting which examples contributed to learning; it's losing track entirely of which inputs shaped the knowledge in the

first place. What remains are judgments, disconnected from their origins, forming the bedrock of human cognition without a clear roadmap of how they were constructed.

Deep learning, one of the most transformative forms of artificial intelligence, operates much like human cognition because it was inspired by this very mechanism of learning by example. Just as children learn through exposure, deep learning systems are trained using labeled examples of what they are expected to recognize. Over time, these systems develop neural networks capable of identifying new, unseen data. For instance, a machine shown enough pictures of cats can learn to recognize cats it has never encountered before. Yet, just like human intelligence, the process by which deep learning systems make decisions is shrouded in opacity. Once trained, the system retains no record of the inputs it learned from or how those inputs informed its decision-making. This lack of transparency in both human and machine cognition underscores a shared trait: an inability to explain the logic behind decisions.

This opacity in deep learning is known as the “black box problem,” and it presents significant challenges. The fundamental issue lies in our inability to understand how a system arrives at its conclusions. This becomes particularly problematic when deep learning systems produce undesirable outcomes. For example, if an autonomous vehicle fails to stop for a pedestrian and causes an accident, diagnosing the error is an immense challenge. Did the system misinterpret the pedestrian’s presence? Was the situation too novel for the system to handle? Without insight into the system’s “thought process,” identifying and addressing the root cause becomes nearly impossible.

Even if retraining the system with new scenarios could improve performance, the complexity of real-world conditions makes it impossible to anticipate every potential one. Consider scenarios such as sunny weather with light fog, salt-streaked roads after a winter storm, or countless other variations. These infinite permutations make it exceedingly difficult to ensure that a system has seen and learned from every possible condition it might encounter. As a result, questions about the robustness and reliability of deep learning systems remain unresolved. But it is worse than that.

THE DAWN OF MACHINE-TARGETED DECEPTION

Deception has always played a critical role in warfare, from feigned retreats in ancient battles to disinformation campaigns during the Cold War. But as warfare becomes increasingly reliant on AI and machine learning, malicious actors are adapting their strategies to exploit these systems.

Rapid advances in machine learning have brought about technologies capable of analyzing vast amounts of data, classifying information, and even making decisions autonomously. In military applications, these technologies could be embedded in decision support systems, targeting algorithms, and logistics planning. Yet, as reliance on machine learning grows, so do the associated vulnerabilities. Unlike humans, machine learning systems can be manipulated through carefully crafted inputs. These inputs may deceive the system into misinterpreting its environment, leading to catastrophic outcomes—like friendly fire or misdirected troop deployments.

Machine learning systems, while transformative, remain vulnerable to creative forms of manipulation that exploit their reliance on external inputs. For example, facial recognition systems used for security can be deceived by adversaries wearing carefully crafted masks or applying specific patterns of makeup. These subtle changes can confuse the system into misidentifying individuals or failing to recognize them entirely, compromising secure facilities or sensitive operations. Autonomous drones that rely on GPS signals for navigation can be misled by spoofed location data. An adversary could trick the drone into believing it is on course while redirecting it to a location where it could be captured or destroyed. Natural language processing (NLP) systems, which generate or analyze human-like text, are similarly susceptible to manipulation.

Operational decision-making algorithms, such as those used in route planning, are another vulnerable target. By injecting false traffic or weather data, adversaries could trick the system into recommending inefficient or even hazardous routes, delaying the transport of troops or supplies in a combat zone. Predictive maintenance systems, which use machine learning to identify equipment failures before they occur, could be manipulated by falsifying sensor data. This might lead to premature equipment replacements, unnecessary downtime or, worse, failures at critical moments. Environmental monitoring systems, designed to detect chemical or biological threats, could be deceived through the introduction of benign substances that mimic the signatures of dangerous agents, prompting unnecessary evacuations or diverting resources away from actual threats. These diverse examples illustrate how adversaries can exploit machine learning systems, not by breaking into them directly but by subtly corrupting the information these systems rely on, making deception a widespread and versatile threat.

Compounding these risks is the opaque nature of machine learning. While engineers can program a system to learn specific tasks, they cannot fully understand how the system arrives at its conclusions. This lack of transparency makes it nearly impossible to predict how the system might respond to novel or malicious inputs.

Mitigation strategies, though essential, remain underdeveloped. Basic measures, such as requiring human oversight and implementing manual override mechanisms, provide some safeguards. However, these are not sufficient for the increasingly complex and autonomous systems now being developed. More sophisticated methods—like continuous testing, operator training, and formal system verification—are critical to addressing these vulnerabilities.

IMPLICATIONS FOR MILITARY OPERATIONS

The Pentagon has embraced machine learning as a cornerstone of future military operations. Since the establishment of the Algorithmic Warfare Cross Functional Team (also known as Project Maven) in 2017, the Department of Defense has sought to harness AI for tasks like analyzing intelligence data, monitoring threats, and automating decision cycles. This vision is enticing, promising to maintain the U.S. military's technological edge in the face of growing competition from adversaries like China and Russia.

However, the adoption of machine learning also intensifies the arms race between attackers and defenders. Adversaries are actively developing methods to exploit AI vulnerabilities, creating a dynamic in which every new capability brings new risks. The challenge lies in balancing the benefits of automation with the risks of deception. As the military becomes more dependent on these systems, the stakes of a successful attack on machine learning capabilities grow exponentially.

The vulnerabilities in machine learning echo earlier challenges faced in the realm of cybersecurity. Historical events like the 1988 Morris Worm and the subsequent creation of the Computer Emergency Response Team (CERT) highlight how slow responses to emerging threats can have far-reaching consequences. Similarly, the establishment of the Forum of Incident Response and Security Teams (FIRST) in 1990 demonstrated the need for global coordination in addressing cyber threats.

Despite decades of progress in cybersecurity, adversaries remain highly effective at exploiting vulnerabilities. The 2019 Secretary of the Navy Cyber Security Readiness Review noted that countries like China and Russia are conducting large-scale, strategic operations to achieve their objectives. The lessons learned from combating cyber threats must now be applied to machine learning systems, which face a similarly urgent and evolving threat landscape.

To address the escalating risks posed by machine learning vulnerabilities, a proactive and comprehensive strategy is essential. The first step is to acknowledge a critical lesson from the past: treating security as an afterthought is a dangerous mistake. Machine learning systems must be designed with security as a core principle, embedded into their development and deployment processes from the outset. This approach ensures that vulnerabilities are mitigated before they can be exploited, rather than patched in response to an inevitable failure.

The "black box" nature of machine learning systems presents a profound challenge, demanding significant investment in research to address fundamental questions about their security. Understanding how to safeguard these systems in sensitive applications and ensure their safe, reliable operation is not just a technical necessity but a critical step in building trust in the technologies shaping our future.

Collaboration is another cornerstone of this effort. Existing organizations like FIRST must evolve to include expertise specific to the complexities of machine learning security. Equally important is fostering partnerships among the private sector, academia, and government to create a unified front. These sectors bring complementary strengths, and their cooperation is vital to addressing the multifaceted challenges that machine learning vulnerabilities present. Without such collaboration, the gaps in our defenses will only widen.

Proactive defensive measures are critical to staying ahead of adversaries. Red teaming exercises, in which systems are subjected to simulated adversarial attacks, can uncover weaknesses that might otherwise go unnoticed. Confidential reporting mechanisms must also be established to

enable rapid responses to vulnerabilities as they are discovered. These measures not only fortify individual systems but contribute to a culture of vigilance and adaptability.

Finally, integrating security into hardware is an essential layer of defense. Machine learning-specific hardware should be designed to include features such as activity audits and mechanisms to prevent unauthorized access. These innovations protect systems at their most fundamental level, ensuring that even if software defenses fail, the hardware remains resilient.

Together, these strategies form the foundation of a robust defense against the vulnerabilities inherent in machine learning. As these technologies continue to shape critical systems in society, the question is not whether we can protect them, but whether we are willing to prioritize the effort required to do so.

A CALL TO ACTION

The rise of machine learning represents both an opportunity and a challenge. While these technologies promise to revolutionize military operations, they also create unprecedented vulnerabilities. The U.S. cannot afford to approach this new frontier with complacency. Proactive measures, informed by lessons from cybersecurity, are essential to securing machine learning systems. Organizations like the National Institute of Standards and Technology (NIST) must play a central role in establishing enforceable standards for AI reliability and safety. Simultaneously, investment in research and development must prioritize security alongside innovation.

By embracing a "whole-of-nation" approach—uniting industry, academia, and government—the U.S. can mitigate the risks of machine learning while reaping its benefits. The alternative is a future where vulnerabilities outpace capabilities, undermining the promise of these transformative technologies.

The age of machine-targeted deception is here. As machine learning systems take on increasingly vital roles in military operations, they become prime targets for adversaries seeking to exploit their weaknesses. The consequences of failing to address these vulnerabilities are dire, ranging from battlefield losses to a loss of trust in critical systems.

Recognizing and mitigating these risks is not just a technological challenge but a national imperative. A new age of deception demands a new level of vigilance, innovation, and cooperation. The sooner we act, the better prepared we will be to face the challenges ahead.

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